

CS5917 MSc Project Manualⁱ

This manual provides additional information about aspects of the course.

It is noted that some of the information detailed in this manual may not be up to date, so please always follow the information provided in myAberdeen if you have concerns.

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1) Ethics Approval

All research undertaken as part of student dissertations at PGT level should be assessed using the student dissertation ethical review checklist (UG & PG Dissertations Checklist, <http://www.abdn.ac.uk/cops/research/dissertations-670.php>) and should thereafter be passed on to the dissertation supervisor before commencing the dissertation project. If the checklist indicates that formal review is required, then the student should submit the research proposal to the College Ethics Board for consideration.

Prof Ehud Reiter should be contacted for any ethics issues arising from the project.

2) Supervisors & Personal Logs

This section summarises two key aspects of the project: the role of your project supervisor, and maintaining a personal log.

The Role of the Supervisor

Each of you has been assigned one or more members of academic staff to act as your project supervisor(s). Your supervisor will arrange to meet with you at regular intervals, as part of his or her project topic group. The purpose of these meetings is to allow you to discuss with your supervisor any project-related activities, and for you to raise any problems/concerns. Normally, a meeting should consist of three parts: i) what you have done since the previous meeting; ii) what issues you encountered (and need to discuss); iii) what you will do until the next meeting.

If you encounter any difficulties during the project, you should communicate in the first instance with your supervisor, who will then report any serious problems to the programme coordinator. Any issues that cannot be resolved in this way should be directed to either the programme coordinator or the Director of Postgraduate Teaching.

Maintaining a Personal Log

It is essential that you keep your supervisor informed of progress and meet regularly (normally once a week). It is recommended that after each meeting you provide a summary of the discussion, the achievements of the past week and goals for the following week. You should take this seriously, as your notes will help you avoiding going over the same issues again, and, more usefully, when you are writing up your report, you will have the notes to start from. Moreover, your notes also serve as a means by which academic supervisors can monitor your progress, achievements, and any problems.

We also strongly recommend that you keep a personal project log (in a notebook or some other form). You should use your log (or a blog) to keep accurate, timely records of project activities. Remember that in August you may not recall why you made certain decisions back in June!

3) Deliverables

During the project you will be asked to produce deliverables; these are presentations, demos, and documents which show progress. This section summarises those deliverables (in the table below) and provides some background detail.

Project Plan

When embarking on a project of this scale, it is advisable to draw up a plan, setting out what you intend to do, and when you intend to do it. You can then use this plan to check your progress throughout the year. The plan will also ensure that you have started your project in a sensible manner, and that you are proposing to tackle a project that has sufficient content for an MSc degree, but is not too ambitious.

Your project plan should be at most 1,000 words, (roughly 2 A4 pages) and should contain:

- A brief introduction to the project, giving the background and explaining why it is a worthwhile task.
- The main goals, making it clear which goals are central, and which are optional extras;
- Risk assessment, describing likely circumstances in which the project would become unfeasible, and the steps that would be required to recover the project.
- A list of the hardware and software resources required.
- An outline timetable.

Ideally, a plan will include a basic project, which should be fairly straightforward to implement, and which should be completed reasonably early. It should also include more ambitious extensions, or possible developments to be tackled depending on the results of your initial work. A properly planned project should not only come together at the very end of the period.

You must discuss your plan with your supervisor before you submit it.

Submit a final copy to your supervisor, and upload a copy to MyAberdeen too via the Assessments section.

Code Submission

The deadline for submission of your program listings is specified within the MyAberdeen pages.

Demonstration

In the last week before your project submission you will be asked to give a short demonstration (lasting around 30 minutes in total including questions) of your system. The major purpose of this is to allow you to show how your system works, and to demonstrate its capabilities. This demonstration should not be regarded as an alternative to a clear and informative dissertation, but as an aid to the proper appreciation by the examiners of your work, and to allow for examiners to ask you questions. It is important you arrange a time with your allocated markers two weeks before the end of the project. You should not assume that your markers will have read your dissertation by the time you have the demonstration (nor should you assume they won't have read it!)

Dissertation

The deadline for submission of your dissertation is specified in the *Course Timetable* page of MyAberdeen. You will be required to submit an electronic copy as a pdf file. Details of how to submit the electronic copy will be given nearer the final deadline. Further information about the dissertation can be found in a fuller *dissertation* section (Section 6).

Late Penalties

The deadline for submission of the project code is a "soft" one, meaning that there are no penalties if you submit after the deadline. However, we strongly encourage you to meet this first deadline, as the time you may spend working on the code beyond the deadline will be at the expense of working on your dissertation (so in an odd way, you are penalising yourself...).

Late submission of the dissertation, however, does attract a penalty: *Work submitted up to one day late attracts a penalty of 10%, up to one week late, 25%; work handed in more than one week late is marked and returned but is counted as a 'No Paper'.*

4) Submission Guidelines

The overall mark for your project (comprising the demonstration, program and dissertation) contributes to 1/3 of your MSc degree. If you are invited to continue work as an MSc student after the May examinations, your project work, demonstration and dissertation serve to discriminate between the award of an MSc, and the award of a Diploma. You may know what you have done in your project, but the examiners can only judge the quality of work by the description that you give in your dissertation. Make sure that your dissertation accurately and clearly reports on the work you have done.

Re-submission is not normally allowed, so you must get it right first time. You should discuss your dissertation with your supervisor, who will normally require to be shown a plan of the structure of the dissertation and one or two draft chapters some weeks before the submission deadline. Remember that your dissertation has to be of an acceptable standard in terms of content and presentation. However good your basic work has been, however brilliant your ideas have been, the

dissertation cannot be accepted if it is not well written and properly presented. Your supervisor is not responsible for proof-reading your dissertation.

5) Demonstration

In the demonstration week (see the course timetable page for the exact date) you will demonstrate your project to two markers. Your presentation should highlight the goal of your project and its important features or issues that it addresses. You should give an introduction to your project (don't assume your markers will know it or indeed that they have read your report before the demonstration), including why it is interesting or worth doing, how you did it, discuss any issues you encountered during your project and how you solved them. You will be marked on how well you present your project and not on the content of the project itself (although we will take into account subjects which are hard to explain).

As you have limited time to answer questions, you will need to think clearly about what you can get over to your audience in a short time. Do not try to put in too much detail. Even though one diagram may be worth a thousand words, don't try to get too much into the presentation. Someone looking at your demonstration should go away with a clear idea of what it was about, what issues you were tackling, how you set about it and what degree of success you had. Use the preparation of your demonstration as a chance to stand back from the actual day to day work you are doing, and think about the issues you are facing. A good project will have many interesting aspects, but you may not have time to report on all of them, so choose wisely.

When marking the demonstrations, markers will consider the following five questions and how you handle discussion and questions about the project:

- Did you introduce and motivate the project, so that a non-specialist could understand the goals?
- Did you speak clearly and hold our attention while you talked?
- Was the demonstration well-prepared and helpful?
- Did you manage your time well, and stay focused on the topic?
- Did you handle the questions well?

6) Dissertation

The dissertation is your chance to tell the examiners what you have done. If you are successful and are awarded the MSc then it will be kept as a Department document and will be available for all to read. You should write it in such a way that it is technical, interesting, readable and understandable by a wide range of people, not just you and your supervisor. Your final submission must be submitted via MyAberdeen.

The report should be no longer than about 20,000 words (roughly 50 pages, single-space, 12pt font), excluding appendices, and it should contain:

What you have done.

Explain what your project was about, describing both the issues involved and the domain you were working in. The dissertation should be a free standing document; the reader should not need to refer to any other documents, or to the appendices, in order to obtain a good understanding of your

work. You should, however, consider putting detailed domain information (for example) in an appendix.

Why you have done it.

Give reasons why you worked in this project, i.e., explain why the project was interesting, challenging or novel, and relevant to your MSc programme.

How you have done it.

Report on how you carried out the work. This should not be a blow by blow account of every event in the course of your project work, nor a line by line description of the program which implements it, but it should be a description of the methods that you used, the decisions that you took, the results that you achieved and the conclusions that you have drawn. Readers should be given in the dissertation an outline of your system and how it does what it does. This should allow them to understand at a high level how to build a similar system and how to interpret your results. Detailed technical information should be provided in your maintenance manual (see below).

Critical Discussion

You must discuss your work and achievements in the light of what other people have done (existing and related work) and what you might like to have done yourself (as future work). This critical analysis is very important, and it is looked for by the examiners. If you think that you made wrong design decisions, then say so. Developing the ability to appraise work critically, including one's own, is a vital attribute for any scientist. Make sure that you appraise your work in the light of the published literature/related work. In addition, at the end of your project, discuss whether you still consider the techniques used appropriate. If this is not the case, then say so.

Much of the material for your critical analysis will come from your work in evaluating your project. See the *verification, validation and evaluation* sections below for advice on doing this.

Structure of Dissertation

Discuss this with your supervisor. Show him/her a draft structure early on. The dissertation should be divided into chapters with an introduction and a conclusion and as many intermediate chapters as you think appropriate. You should ensure that you outline the background to your project, discuss aims and objectives and so on. The report can also have appendices. Remember that many of your readers will want to be able to understand the broad issues without necessarily going into the fine details. Material which is essential to the deep understanding of your work (for example, detailed accounts of experimental runs), but not essential to the general reader should be put in appendices.

You should include an abstract, acknowledgements, references, and a signed declaration that the work is your own. You must include the following declaration in your dissertation:

No portion of the work contained in this document has been submitted in support of an application for a degree or qualification of this or any other university or other institution of learning. All verbatim extracts have been distinguished by quotation marks, and all sources of information have been specifically acknowledged.

User Manual

Where appropriate, you should supply a user manual, so that someone other than yourself, or your supervisor, is able to use your system/program and its functionalities, change settings and parameters, and replicate (if applicable) your experiments and results. It is a good idea to find a friend and get them to test the user manual by trying to run the system from its instructions. The user manual is usually provided as an appendix of the dissertation (and thus will not count towards the word/page limit).

Maintenance Manual

This should be used to describe the details of your implementation. It should be usable by people wanting:

1. To install the program;
2. To modify your program (e.g. for a different domain);
3. To extend the program;
4. To trace bugs in its execution.

This is an important part of your project documentation, and you should ensure that you include details such as:

- Instructions on how to install the system (From the compressed archive you submit);
- Instructions on how to compile/build the system;
- Hardware/software dependencies (Including libraries, other software packages used);
- Organisation of system files (Including directory structures, location of files within various directories, details of any temporary files created);
- List of source code files (With brief summary of role in system);
- Crucial named constants (Document constants to change when modifying the system, e.g. array size.)
- File pathnames (Explain your philosophy for accessing files of data or parameter values. Are the file pathnames given on command line or in environment variables or embedded in program text or as named constants?)
- Modifications (Include, if relevant, suggestions as to what needs to be altered if the system were to be extended - as described in the future work section of your report).

The maintenance manual is usually an appendix of the dissertation.

7) Project Code

You should prepare code submission on the date specified on the Course Timetable page in MyAberdeen.

Program Listings

You should prepare a PDF of your code as detailed on the 'information' page containing a listing of your programs together with any test programs, data and results. Remember that the information must be readable. You do not need to print out your code. Remember to label the listings clearly so that one knows what the program is, what test program(s) is/are, etc. Please make sure you include a contents list.

Archiving your Project

The Department retains archives of all projects for possible demonstration and future use. As part of your project submission you are asked to provide the software you have developed in a format that can be used by others. To this end, you should provide a compressed archive file containing the following:

- A one page description of the program;
- Installation instructions - what does a user have to do to get your program running on their machine? How do you compile the program?
- Dependencies - what other elements does the program require to run? Examples include: hardware requirements, operating systems, software packages, libraries. Include version numbers.
- Source code - make sure that you use relative pathnames rather than absolute ones, so that the system can run without depending on your login name.
- Executable version of the code(s), where applicable.

Details on how to submit your project archive will be available nearer the deadline.

Note that ensuring that your program can be installed and used is a vital part of software development. The instructions and code you provide will be tested during the project demonstration.

8) Method of Assessment

Who Marks the Project?

The project is marked by two members of staff, neither of whom will be your supervisor. Both mark the project independently and give their marking form to one of our secretaries, for lodging their marks. After both have marked the project, then they meet to discuss and agree on a joint mark. If the two of them came to different marks, then they need to explain how and why they came to an agreed mark. If there is a large disagreement, and the two markers cannot agree on a mark, then the matter will be adjudicated by a third party. An external examiner looks at all the dissertations and marking forms, for quality control.

Using two markers, neither of whom was the project supervisor, provides a more robust method of assessment. This is adopted by many universities to reduce positive or negative bias on the part of the supervisor.

Given that neither project marker is going to be your supervisor, then it is important that you do not assume that readers will necessarily know what your project is about. This means there is more responsibility on you to be clear, concise, technical, and provide evidence of breadth and depth of knowledge in specific sub-areas, as well as techniques, technologies and research methodologies. If your project has an associated system which you developed, then you must make sure that you report on the choices that you made in the project, reasons as to why developed it one way (and what other options were and why they were not used instead), problems you encountered and how you overcame them, how you tested and evaluated your system/ideas, and so on. Your supervisor would know these issues, but the people marking your project may not. It is important that you report on these issues for the benefit of the markers (and ultimately, your benefit).

Marking Scheme

Please refer to the marking form available in MyAberdeen. Please make sure you read the marking scheme before and during your project, to ensure you are working on what will be assessed.

9) Intellectual Property Rights

According to current law/regulations in place in the UK/EU, software produced by students who are not employees of the University but are often in receipt of a government grant, is treated in the same way as if they had written a novel at home in the evenings. In consequence, where you are actually producing software as part of your regular daytime activity under the direction of a University supervisor, and using University computers and software tools/compilers, we need to establish clear title to the IPR. Hence, since 1994, the University Court takes the act of registration to include assignment of IPR to the University.

The reason for this is not so much that the University can get lots of extra income, which is unlikely, but to defend its ability to do research. Research groups must be able to control the copyright of software they are developing, in order that future staff and students can continue to use and adapt it without threat of legal action or predatory pricing. One should also remember that although your piece of software seems very valuable to you, it usually only works as part of a larger system built by several people either as a team or in succession.

Note that you have not signed away your birth right. The agreement only concerns IPR “arising in connection with or incidental to your course and studies here”. You have not signed away rights to software developed at home for your own purposes on your PC. Instead you have entrusted the University with your claim to the IPR developed here in exactly the same way that University paid research and teaching staff have to do, so that any IPR can be safeguarded, and any financial reward shared out on a fair and equal basis.

In the case of students undertaking industrial projects as part of their course, we ask them to sign a form which also covers arrangements for examining reports and publication rights. The form assigns IPR to the University, and not the collaborating body. If the University makes money from the software, this will be shared equitably with you. For further information, contact the Department Industrial Liaison Officer.

10) Testing and Evaluation

There are different types of projects, so you should discuss with your supervisor on how to perform evaluation and testing for your specific types of projects.

ⁱ Date Created: 16 March 2018 by Dr Wei Pang and reviewed by Prof Wamberto Vasconcelos.